

3ai	work done per unit mass bringing (small test) mass from infinity (to the point)	E	B1
3aii	(near Earth's surface change in) height $\ll$ radius or height much less than radius potential inversely proportional to radius and radius approximately constant (so potential approximately constant)	A	B1 B1
3b	curve from r to $4r$, with gradient of decreasing magnitude and starting at $(r, \pm\phi)$ <u>and</u> line passing through $(2r, \pm0.5\phi)$ and $(4r, \pm0.25\phi)$ line showing potential is negative throughout	A	B1 B1
3ci	Gain in KE = loss in GPE $\frac{1}{2}mv^2 = 0 - (-GMm/R)$ At distance $R = 3r$, $v = \sqrt{\frac{2GM}{3r}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{3 \times 6.4 \times 10^6}} = 6.46 \times 10^3 \text{ m s}^{-1}$ At distance $R = 4r$, $v = \sqrt{\frac{2GM}{4r}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{4 \times 6.4 \times 10^6}} = 5.59 \times 10^3 \text{ m s}^{-1}$ Change in speed = $6.46 \times 10^3 - 5.59 \times 10^3 = 8.7 \times 10^2 \text{ m s}^{-1}$	D	C1 C1 A1